



SCHOOL OF COMPUTING SCIENCE AND INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
PROGRAMME: MASTER OF INFORMATION TECHNOLOGY

Course Code & Course Title : 05BMSIT23161: OBJECT ORIENTED PROGRAMMING - CORE JAVA

Course Frame Work

Credits: L-T-P: 4 - 0 - 2

Total Credits: 4 + 2

Contact Hours/Week: 5

Direct Teaching Hours: 60

Total Contact Hours: 60 + 30

Course Learning Objectives:

The course aims to understand the basic concepts such as

- Understand the fundamental concepts of object-oriented programming and the evolution of Java.
- Develop and execute basic Java programs using proper syntax, structure, and environment setup.
- Apply object-oriented principles such as class. Objects, methods, constructors, inheritance, and polymorphism, interfaces, packages, multithreading, exception handling and files.
- Build graphical user interfaces using JavaFX components and layout managers, and apply event handling and styling.
- Integrate Java applications with databases using JDBC for real-time problem-solving.
- Real-World Application Development

Course Outcomes: On completion of the course, student would be able to:

CO's	Course Outcomes
CO1	Understand the Fundamentals of Java and Develop Basic Java Programs Using Core Constructs like Variables, Data Types, Operators, Decision Making and Looping.
CO2	Utilize Object-Oriented Programming Concepts and Work with Classes, Objects and Methods, Arrays, Strings, Vectors, Wrapper Classes.
CO3	Develop Programs Using Advanced Java Features and Implement Inheritance, Interfaces, Package, Multithread, Exception Handling Concepts.
CO4	Design and Execute GUI Programming with JDBC and work with real word problems.
CO5	Build Problem-Solving Skills Using Java.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	0	0	0	0	0	0	0
CO2	3	3	3	2	2	0	0	0	0	0	0	0
CO3	2	3	3	2	2	0	0	0	0	0	0	0
CO4	2	2	2	2	3	0	0	0	0	0	0	0
CO5	3	3	3	3	3	0	0	0	0	0	0	0

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	2	2	2	2	0
CO2	3	3	3	2	2	0
CO3	2	3	3	2	2	0
CO4	2	2	2	2	3	0
CO5	3	3	3	3	3	0

	>70-100 High	>40 & <70 Medium	<40 Low	
	3	2	1	
Syllabus	Teaching Methodology			Hours
UNIT I				
Object Oriented Programming 1.1. Introduction, Concepts, Benefits, and Applications. Java Evolution: Java History, Java Features, Differs from C and C++, Java Internet and www, Java Support Systems, Java Environment. 1.2 overview: Simple Java program, Java Program Structure, Java Tokens, installing configuring Java, implementing Java Program, Java Virtual Machine, Command Line Arguments. Constants, Variables, Data Types: Declaration of Variables, Giving Values to Variables, Scope of Variables, Symbolic Constants, Type Casting, Getting Values of Variables, Standard Default Values. 1.3 Operators and Expressions: Arithmetic Operators, Relational Operators Logical Operators, Assignment Operators, Increment and Decrements Operators, Conditional Operators, Bitwise Operators, Special Operators, Arithmetic Expressions, Evaluation of Expressions, Type Conversion in Expressions, Precedence of Arithmetic Operators, Mathematical Functions. 1.4 Branching: Decision Making with if Statement, Simple if Statement, if.....else Statement, Nesting of if.....Else Statements, else if Ladder, Switch Statement? Operator. Looping: while Statement, do...while Statement, for Statement, Jumps in Loops, and return statements, Labeled Loops.	• Use multimedia presentations, whiteboard explanations. • Real-world examples programs			15
UNIT II				
2.1 Classes, Objects and Methods: Defining a Class, Declaring Variables and Methods, Creating Objects, Accessing Class Members, Constructors, Static Members, Nesting of Methods. 2.2 Arrays, Strings and Vectors: One-dimensional Arrays, Two -Dimensional Arrays, Strings, Vectors, Wrapper Classes.	• Use multimedia presentations, whiteboard explanations. • Real-world examples programs			15

2.3 Inheritance: Extending Classes, Creating Single, Multilevel, Multiple, Hierarchy and Hybrid, Final Variables and Methods, Finalizer methods, Abstract Methods and Classes, Visibility Control. Polymorphism: Method Overloading, Method Overriding, Dynamic Method Binding. 2.4 Interfaces Defining Interfaces, Extending Interfaces, Implementing Interfaces, Implementing multiple Inheritance by using Interface. Accessing Interface Variable.		
UNIT III		
3.1 Packages: Java API Packages, Using System Packages, Naming Conventions, Creating Packages, accessing a Package, using a Package, adding a Class to a Package, Hiding Classes. 3.2 Multithreaded Programming: Creating Threads, Extending the Thread Class, Stopping and Blocking a thread, Life Cycle of a thread, Using Thread Methods, Thread Exceptions, Thread Priority, Synchronization, Implementing the 'Runnable' Interface. 3.3 Exception Handling: Types of Errors, Exceptions, Multiple catch, Nested Try Catch, Throw, Throws, Finally and Built-in exceptions, Throwing our own exceptions. 3.4 Files: Concept, Stream Classes, Byte streams classes, Character stream classes, File class, Creation of files, Reading Writing Bytes, Reading Writing Characters, Concatenating and buffering files. Random Access Files.	• Use multimedia presentations, whiteboard explanations. • Real-world examples programs	15
UNIT IV		
Java FX: 4.1 Introduction: What is JavaFX? JavaFX vs Swing, Setting up JavaFX in an IDE, JavaFX application lifecycle. Basic JavaFX Structure: Application class, Stage and Scene, Layout containers overview. 4.2 GUI Controls (Basic Components): Label, Button Text Field, Text Area, Password Field. Layout Managers (Panels): VBox, HBox, BorderPane, Grid Pane, Stack Pane, Flow Pane. Event Handling: Handling button clicks, Event	• Use multimedia presentations, whiteboard explanations. • Real-world examples programs	15

Handler interface, Lambda expressions in JavaFX, Handling keyboard and mouse events.

4.3 Intermediate UI Components: CheckBox, RadioButto, ToggleGroup, ComboBox, ListView, TableView. **Styling JavaFX Applications:** Using CSS in JavaFX, Inline styles vs external style sheets. **Media and Charts:** Displaying images, videos and charts.

4.4 JDBC Java Database Connectivity, Mini Project / Case Study: Real-time problems.

Text Books:

- 1) E Balagurusamy, "Programming with JAVA", McGraw Hill 7th Edition, 2023.
- 2) Herbert Schildt, Dr Coward, Danny, "Java-The Complete Reference", McGraw Hill, 11 Edition, 2024.

Reference Books:

- 1) Ken Arnold, James Gosling "The Java TM Programming Language", Addison- Wesley, 2006.
- 2) C. Xavier, "Programming with JAVA 2", Scitech Publications (India) Pvt Ltd, 2008.
- 3) Santhosh, "Spring and Hibernate", Tata McGraw-Hill, 2009.

Web References:

- 1) <https://beginnersbook.com/java-tutorial-for-beginners-with-examples>
- 2) <https://docs.oracle.com/javase/tutorial>

Activity Based Learning/Practical Based Learning:

- 1) <https://www.infosys.com/about/springboard.html>
- 2) <https://www.coursera.org/courses?query=java>

Beyond Syllabus

- **Advanced Java Features:** Understanding bounded and wildcard types.
- **Modern Frameworks and Tools:** Learn Spring Boot for building micro services and web applications.
- **System Design and Architecture:** Implement patterns like Singleton, Factory, Builder & Observer.
- **Performance and Optimization:** Study garbage collection, JVM tuning, and memory management.
- **Testing and Quality Assurance:** Write unit and integration tests
- **Cloud and DevOps Integration:** Use Docker to containerize Java applications.
- **Emerging Topics:** Explore frameworks like Project Reactor for building asynchronous applications.
- **Real-World Applications. :** Contribute to open-source projects to gain practical experience.

Experiments

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Exercises Nos.	Experiments
Demonstration 1	Write a program to find area of geometrical figures using method.
Demonstration 2	Write a program to generate factorial, Prime, Fibonacci of list of number and reading input as command line argument
Demonstration 3	- Design a class to represent a bank account include the following members: - Account Number, Name of the Depositor, Balance amount, to assign initial values - To deposit an amount, To withdraw an amount after checking balance, To display the details.
Demonstration 4	Write a program to implement Methods, and Constructor concepts using OOP Java
Demonstration 5	Write a program to implement Overloading and Arrays concepts using OOP Java
Demonstration 6	Write a program to implement, Strings, Vectors concepts using OOP Java
Demonstration 7	Write a program to implement Package and Multithread concepts using OOP Java
Demonstration 8	Write a program to implement Inheritance and Polymorphism concepts using OOP Java
Demonstration 9	Write a program to implement Interfaces and Exception Handling concepts using OOP Java
Demonstration 10	Write a program to implement Files and GUI concepts using OOP Java
Demonstration 11	Case study on Current Problems
Demonstration 12	Mini project: Real-time Problems using Java, JavaFX and JDBC.